

Probabilistic Hierarchical Mesh Graph Networks for One-to-Many Engineering Simulation

Abstract

Learned mesh-based simulators are trained under a pointwise regression loss and therefore estimate a conditional mean. When a latent field governs the response but is invisible to the model—an unmeasured geometric imperfection, a process-to-process manufacturing deviation—the conditional law of the field is broad and its mean need not be a physically realisable state at all. We develop two mesh-native probabilistic extensions of Hierarchical Interpolating MeshGraphNets that share one backbone and differ only in where the stochasticity is placed: MeshGraphNets-V, a conditional variational simulator with per-level graph latents and a flow-matching conditional prior, and cHI-MGNflow, which learns a conditional flow directly on the field. On industrial warpage data MeshGraphNets-V outperforms cHI-MGNflow by $2.2\times$ on normalised 1-Wasserstein distance with no overlap across an eight-arm design each, but the ranking is the least informative result: decomposing the error shows two orthogonal failure modes, MeshGraphNets-V correctly centred and roughly $2\times$ too narrow, cHI-MGNflow failing on one part family through an almost pure mean offset whose sign flips between families. Neither model is calibrated—the best dispersion ratio attained by either is 0.769 against a target of 1—which is disqualifying for tolerance analysis and is not visible in the ranking. We further show analytically that fitting a flow-matching prior to a jointly trained encoder does not supply a variational rate term: the optimal loss floor is linear in the trainable target scale, so the objective can always be reduced by shrinking the target, while the corresponding KL divergence is zero at every scale. Measured on this task its net effect is nil, which rules the coupling out as a remedy without explaining the deficit. Every claim is bounded by the same limitation—each evaluation set holds one geometry, so a failure to transfer a conditional distribution cannot be separated from a failure on particular parts—and no industrial dataset resolves it, because the withheld variable is never known. We therefore construct a benchmark of axially compressed shells in which it is: a specified band-limited imperfection field of RMS $10^{-3}t$, some $150\times$ smaller than the signature the model can see, yet fully determining which post-buckled branch the shell reaches. Characterisation before any training confirms the required properties and yields a result that reaches back into the first study: the discrete mode label separates only half the geometry pairs that the field separates, so any evaluation resting on a scalar summary measures less than the data contains.

1 Introduction

1.1 From point predictions to distributions on meshes

Graph-based learned simulators represent a discretised physical system by nodes, edges and field values. Graph Network-based Simulators established an encode–process–decode architecture for particle dynamics, and MeshGraphNets adapted this approach to simulation meshes [37, 44]. These models learn point predictions using squared regression losses. Input noise used to stabilise their rollouts should be distinguished from an explicit model of the predictive distribution.

Hierarchical processors extend the spatial range of message passing. MultiScale MeshGraphNets uses information at multiple mesh resolutions, while BSMS-GNN constructs coarse graphs by bi-stride pooling [5, 9]. Operator architectures such as FNO, GINO and Transolver provide other ways to represent nonlocal dependencies [21, 22, 49]. Architecture and prediction target are separate choices: using a hierarchy does not by itself turn a point regressor into a probabilistic simulator. Conversely, hierarchical probabilistic graph models already exist in weather forecasting and fluid simulation [23, 24, 36].

1.2 What a squared loss elicits

For finite second moments, minimising expected squared error elicits the conditional mean. Absolute error elicits a median [11]. The mean can be useful even when the conditional distribution is broad, but it does not specify its spread, tails or dependence. In a multimodal problem, the mean can also lie between physically admissible branches. Thus, an accurate mean alone does not establish an accurate distribution.

GenCFD examines this distinction for fluid statistics and contrasts mean-square-trained deterministic models with generative models [33]. Related observations about averaging incompatible futures occur in stochastic video prediction [3, 6, 19]. For engineering meshes, the practical question is whether a surrogate can generate plausible fields with the correct conditional frequencies, including the distribution of quantities used in tolerance or reliability calculations.

1.3 Uncertainty is relative to the observed inputs

Let g denote the information supplied to a model. Several different mechanisms can produce variation across samples, and their interpretation depends on which variables are included in g .

Chaotic dynamics. For a deterministic solver conditioned on its complete state and parameters, the next state is a point mass. Varying initial conditions produces a trajectory ensemble when those conditions are marginalised out. Partial observation, coarse resolution and model error can introduce additional predictive uncertainty; it would therefore be too strong to identify all weather or turbulent-flow uncertainty with perturbations of a fully observed state [16, 40].

A withheld physical variable. A deterministic solver can produce different outcomes at the same visible condition when an imperfection or material field is omitted. The induced law is $p(y \mid g)$, obtained by integrating over that field. The spread is irreducible for the stipulated input set, but may decrease if additional measurements are supplied. This does not, by itself, classify all physical variability as epistemic uncertainty: data variability and uncertainty about the predictive model are different notions [14]. The controlled shell benchmark in this work uses a specified random imperfection as the withheld variable.

An intrinsically stochastic simulation. A kinetic Monte Carlo process can vary even at fixed physical inputs because of its random draws [10]. Fixing all visible conditions gives a single distribution to learn. Such a task can quantify uncertainty at that condition, although it cannot test transfer between conditions. For grain microstructures, arbitrary grain identifiers also require a representation that respects relabelling symmetry.

1.4 What an evaluation can establish

An RMSE of an ensemble mean evaluates the mean. It cannot identify the ensemble’s dispersion or spatial dependence: ensembles with the same mean receive the same value. A proper score, in contrast, evaluates a predictive distribution against an observed outcome. One outcome per condition is enough to form an unbiased observation of the conditional expected score; averaging over test cases can compare predictive procedures. Repeated outcomes at the same condition are valuable for estimating that particular conditional law without pooling assumptions. Their absence limits condition-specific diagnostics, not the validity of proper scoring [11].

Finite model ensembles introduce a separate estimation issue. If members are independent draws from a predictive law, the empirical-distribution CRPS has a finite-ensemble term that can favour under-dispersed sampling laws. Fair estimators remove that term under their sampling assumptions [8, 50]. The number of reference outcomes and the number of generated members play different roles and must be reported separately.

Propriety is an expected-score property, not a guarantee of a correct ranking in every finite test. Multivariate energy scores may have limited power against some dependence errors, motivating complementary variogram scores and diagnostics of spatial structure [31, 38, 45]. Rank histograms and spread diagnostics help interpret calibration, but cannot establish joint calibration alone [12]. Section 5 separates these statistical properties from the metrics actually available in the reported experiments.

1.5 Contributions and evidence scope

1. **Two implemented probabilistic mesh simulators.** MeshGraphNets-V uses graph-level latent slots across hierarchy levels, aggregate-posterior MMD regularisation and a conditional flow-matching prior. cHI-MGNflow performs conditional flow matching in field space with a hierarchical graph processor. Section 4 documents the implemented objectives and sampling paths. These are related constructions, but their training objectives and sampling costs differ, so their comparison is not a controlled ablation of stochasticity placement alone.
2. **A recorded industrial warpage comparison.** The retained research note reports lower normalised scalar Wasserstein distance for MeshGraphNets-V and under-dispersion in both model families on three evaluated parts. Section 6.1 reports the numerical results together with the missing historical configurations and raw-output limitation. This evidence concerns a peak-to-valley quantity of interest, rather than validation of a joint field distribution.
3. **An analytical check of a trainable flow-matching target.** A solvable Gaussian example shows that the optimal flow-matching regression loss can decrease as the encoded target contracts even when the prior matches the target distribution exactly (Section 4.5). This distinguishes the regression loss from a variational rate term. It does not prove that the complete regularised simulator collapses or explain its observed calibration error.
4. **A controlled shell benchmark implementation.** The generation plan contains 22 geometries in four tiers, with a visible deterministic imperfection and a separate random component withheld from model inputs. Section 6.2 distinguishes planned, solved and retained samples and reports dataset characterisation. No trained-model result on this benchmark is claimed.
5. **An evaluation protocol and a dataset-contract audit.** Sections 5 and 3 specify fair ensemble scoring, diagnostic controls and the information needed to define a one-to-many task. The audit identifies observed initial-state variation in converted datasets and the failure of

raw grain-label regression to respect relabelling. Protocol definitions are distinguished from completed model evaluations.

2 Related work

2.1 Generative PDE surrogates

Generative PDE models address several targets, including accurate temporal evolution, stable long rollouts and distributions of physical states. PDE-Refiner uses denoising refinements to improve the representation of low-amplitude spatial frequencies in neural PDE solvers [26]. Autoregressive conditional diffusion has also been studied for incompressible and transonic flows and turbulence, with attention to temporal stability and spectral behaviour [16]. GenCFD explicitly targets statistical fluid quantities [33]. These objectives should be compared using their stated observables: pointwise rollout accuracy, stationary statistics and conditional probabilistic accuracy are related but distinct.

2.2 Flow matching and coupling choices

Flow matching trains a time-dependent velocity through regression on a conditional probability path without integrating the learned ODE during the basic training objective [25]. Rectified flows and stochastic interpolants provide closely related constructions [1, 2, 27]. The absence of an ODE solve in that objective makes joint training with an encoder practical, but does not make its full cost identical to an ordinary regression head; generation still requires numerical integration unless a different sampler is supplied.

Noise–data coupling is a modelling choice. Minibatch optimal transport can reduce path length and improve flow-matching efficiency, but a finite-batch coupling should not be identified with exact population optimal transport [39, 47]. These literature variants provide context; the field-flow implementation described here pairs Gaussian noise with each training target independently.

2.3 Generative models on unstructured domains

Diffusion Graph Networks and their latent-space variant use multiscale GNNs to sample statistically developed flows on unstructured meshes, including varying wing geometries. Their experiments explicitly compare multiscale and single-scale processors [24]. The subsequent scale-autoregressive model generates coarse mesh fields first and finer fields conditioned on those predictions, using a hierarchy derived from the mesh and a flow-matching sampler [23]. These are direct precedents for hierarchical probabilistic mesh generation.

FluidFlow applies conditional flow matching on unstructured meshes [42]; the Generative Latent Neural PDE Solver maps different meshes to a shared structured latent grid before latent flow matching [20]. The present implementation instead retains graph processing for its latent conditioning and field generation. Differences in representation and training factorisation motivate comparison, but do not establish superiority without matched experiments.

2.4 Latent-variable surrogates and posterior collapse

Conditional VAEs learn a recognition distribution that sees the training target and a prior used when that target is absent [46]. Mismatch between the two distributions can affect generation even when posterior reconstruction is accurate. Learned conditional priors and regularisation address

this problem without guaranteeing its elimination [6, 51]. The MMD-based aggregate-posterior regulariser used here is motivated by InfoVAE.

Lucas et al. analyse posterior collapse through linear VAEs and probabilistic PCA, showing that the linear ELBO need not introduce additional spurious local maxima relative to the marginal likelihood [28]. This does not establish that every nonlinear conditional VAE must collapse or that KL annealing is ineffective. Similarly, a per-example KL term can constrain the aggregate posterior as well as information carried by the latent; the difference from MMD is not the complete absence of aggregate control. Our Gaussian calculation concerns the scale sensitivity of an additional flow-matching loss, rather than a general theorem about VAE training.

2.5 Probabilistic weather forecasting

WeatherBench 2 includes probabilistic verification alongside deterministic forecast metrics [43]. GenCast learns an ensemble forecast distribution, while AIFS-CRPS uses an almost-fair CRPS training objective [18, 40]. Their verification against one realised weather trajectory per forecast case is a valid use of proper scores, with statistical conclusions drawn across cases.

Graph-EFM is especially close architecturally: it samples coarse latent representations in a hierarchical graph, uses a variational objective, and adds an unbiased two-member marginal CRPS term during fine-tuning to improve calibration. It also discusses the tension between marginal scoring and spatial coherence [36]. We therefore do not claim hierarchical stochasticity, proper-score training or this coherence issue as new. The implementation studied here uses per-sample engineering meshes, aggregate MMD regularisation and a flow-matching prior; its empirical evidence must still be judged independently of those design choices.

2.6 Operator uncertainty and microstructure

Bayesian operator methods, deep ensembles and conformal methods address different aspects of uncertainty [17, 29, 30, 41]. Posterior parameter draws can induce coherent function samples; it would be incorrect to describe every such method as producing intervals only. Coverage guarantees and joint generative accuracy should also be distinguished. Probabilistic neural operators provide another route to distributional learning using scoring-rule objectives [4].

For microstructures, grain identifiers are arbitrary, whereas grain boundaries, sizes and correlation functions represent physical or morphological structure [10, 48]. Generative methods evaluate such structure using descriptors and statistical comparisons [13, 15, 34]. Marginal scores on invariant boundary indicators remain meaningful for local event probabilities, but do not determine joint morphology. Conversely, matching a few descriptor moments does not establish equality of field distributions. These complementary limitations motivate reporting both the representation being scored and the information it discards.

2.7 Positioning

The contribution is a documented implementation and an evidence-bounded engineering study combining probabilistic graph models with an explicit hidden-variable benchmark construction. Hierarchical generation, learned latent priors and proper scoring all have prior art. The recorded industrial comparison evaluates a scalar response on three parts; the shell study characterises a generated dataset before model training. Neither establishes that our hierarchy improves joint field structure, that a persistent latent has been learned across a rollout, or that calibrated transfer to new shell families has been achieved. Those claims require additional experiments.

Table 1: The partition of failure modes. No surveyed dataset has all three of (a) an unstructured mesh with nodal fields, (b) Monte-Carlo replicates at a fixed condition, and (c) many distinct conditions.

Has	Missing	Exemplar
Replicates + many conditions	unstructured mesh	The Well (all subsets), PBFM, StocBench
Replicates + real CAE mesh	many conditions	SimJEB (381×1 condition)
Many conditions + real CAE mesh	replicates	DrivAerNet++, CFDBench, Mechanical MNIST
Replicates + real CAE physics	the mesh itself	Airfoil RANS ($25 \times 5,000$, regrid-ded to Cartesian)
Many ground-truth realisations	condition diversity	CESM2 LENS2 (100 truths, ~ 2 conditions)

3 The benchmark landscape for one-to-many simulation

3.1 The admission criterion

A dataset can validate a distributional claim only if it supplies *multiple ground-truth realisations at a single model-visible condition*. This is a harder requirement than it sounds, and most datasets described as “stochastic” in the neural-PDE literature fail it. Chaos is not sufficient: a chaotic system with one trajectory per parameter setting still gives one draw from the conditional, and a distance to one draw is not a divergence between distributions. Nor is visual variability sufficient. We surveyed 87 candidate datasets across curated PDE suites, weather and climate ensembles, turbulence archives, solid-mechanics and fracture sets, engineering CAE collections, stochastic-PDE benchmarks and adjacent generative domains, and classified each by three properties: replicate count at a fixed condition, the representation (mesh, graph, or structured grid), and *what is hidden from the model* that creates the spread.

Table 1 summarises the outcome. The organising finding is negative and, we argue, publishable in itself.

Weather supplies *many conditions $\times$ one truth*; large climate initial-condition ensembles supply *many truths $\times$ two conditions*. Nobody has built many conditions $\times$ many truths on a mesh.

3.2 What is admissible today

Only a handful of obtainable datasets clear the criterion with a replicate count adequate for a fair-CRPS estimate, and four of them belong to a single collection — The Well [35] — which is the only corpus that systematically ships seed ensembles.

Turbulence-gravity-cooling (100 seeds $\times$ 27 conditions). The only combination in the survey supporting *both* a stable per-condition CRPS and rank histogram *and* a held-out-condition calibration test. This is the headline conditional result. Its cost is storage (~ 829 GB) and a free-fall-time-normalised cadence that must be reconciled before pairing seeds.

Rayleigh–Bénard (10 seeds $\times$ 35 conditions). Chebyshev-graded wall-normal spacing makes it the most mesh-like structured set available and mildly mis-specifies uniform-grid CNN baselines. Ten replicates suffice for CRPS but are thin for rank histograms. At high Rayleigh and Prandtl number the spread collapses, so per-condition spread must be reported rather than a pooled number.

Turbulent radiative layer 2D (10 seeds $\times$ 9 conditions). At 6.9 GB this is the ablation harness, not a result: nearly one-dimensional conditioning makes it a strong calibration test and a weak expressivity test, and 90 trajectories is a small training set for a generative mesh model.

Viscoelastic instability (four coexisting attractors). The only genuinely *multimodal-with-discrete-modes* dataset available, and therefore the direct empirical answer to “why a flow-matching prior instead of a heteroscedastic Gaussian head”: a unimodal head provably cannot represent four attractors. It is unconditional, so conditional calibration is untestable here, and unequal mode counts and trajectory lengths must be reweighted explicitly or a sampler will confuse mode weight with sampling artefact.

Airfoil RANS ensembles. The only entry whose uncertainty mechanism — solver non-convergence plus turbulence-closure deficiency — is the one a CAE practitioner actually meets, and the only one whose conditioning variable *is a geometry*. It is distributed regridded to a Cartesian lattice, so exercising a mesh hierarchy requires re-running the cases or obtaining the raw mesh.

3.3 Instructive rejections

Several datasets are cited in probabilistic work as though they satisfied the criterion. They do not, and the reasons are worth naming.

- **One realisation per model-visible condition.** PDEBench and PDEArena sample the governing parameter continuously, so no two runs share a condition; the ACDM scenarios ship one trajectory per Reynolds or Mach number, which is why their “posterior sampling” is scored against physics statistics rather than a ground-truth divergence [16]; JHTDB stores exactly one DNS per configuration; DrivAerNet++ one CFD run per geometry [7].
- **Temporally correlated snapshots posed as an ensemble.** Several archives present N snapshots of *one* trajectory as N samples. That is a decorrelation-time question, not an uncertainty question, and it is the most common silent error in this area.
- **Averaged targets.** Where a benchmark releases the *mean* of a small initial-condition ensemble as its target, an emulator wins by suppressing variability, and RMSE against a three-member mean is substantially scoring noise.
- **Model-form variation mistaken for aleatory spread.** Sets whose “repeats” are alternative energy decompositions, alternative constitutive models, or *neural-network training seeds* are varying model form, not the physical law’s randomness. Pooling them into a spread estimate is a category error.

The most instructive rejection is the canonical near-miss. Mechanical MNIST Crack Path exhibits dramatic path sensitivity under an identical loading protocol, and is regularly invoked as evidence of stochastic behaviour. But as published, the realised inclusion centres are handed to the network as a 64×64 rigidity array and the resulting map is solved deterministically at $F_1 = 0.870$ [32]. *Sensitivity is not uncertainty*. Three separate recent fracture and damage benchmarks shipped with visually dramatic stochastic-looking behaviour and zero aleatory replicates. Naming this confusion, we think, is overdue.

3.4 A withholding construction, and its measured information floor

This subsection and Section 6.2 describe *two different* datasets, reached by the two routes available when no admissible benchmark exists. Here we convert an existing dataset by withholding a variable it already contains; there we generate one whose withheld variable is specified in advance. The contrast is instructive, and the reason the second route was also needed is measured below: a converted dataset’s withholding is never perfect, because the hidden variable partially images itself into the visible state. Only the buckling benchmark of Section 6.2 supports the transfer question this paper’s experiments leave open, since it alone supplies many geometries with many realisations each.

We turn the canonical near-miss into an admissible benchmark by an explicit, declared act of information withholding. The rigid-inclusion field is *not* converted; the model sees reference coordinates, the damage and displacement state, and nothing else. Every case then presents a bit-identical visible input, so the released samples are draws from a single conditional and the empirical conditional distribution is directly observable.

We report four properties measured on the converted data, because a withholding construction is only defensible if its consequences are stated rather than assumed.

The spread is large and continuous. Pairwise relative L^2 of the damage field rises from 0.047 at the first stored step to 0.948 at the last; final-crack IoU averages 0.283 with a 95th percentile of 0.473. A PCA of the final damage fields requires 43 components for 90% of variance, and no sample has a nearest-neighbour crack IoU above 0.9. There is no discrete mode structure: the outcome distribution is continuous and high-dimensional, which is precisely why a descriptor-histogram acceptance test would be inadequate and a proper score is required.

The latent is structured, not i.i.d. Inclusion centres are seeded on bitmap pixels above an intensity threshold, with sub-pixel jitter and a minimum-spacing rejection rule. The hidden variable is therefore a structured, class-clustered prior with a genuinely random realisation drawn on top of it — not white noise. Any paper using this construction should say so; the distinction changes what a well-calibrated prior is expected to look like.

The withholding is not perfect, and we quantify the leak. The first stored frame is not the pristine notch but an early loading step, so the recorded displacements already encode the *elastic response* to the hidden inclusions. Nearest-neighbour retrieval on that first frame predicts the final crack at IoU 0.2726 against a random baseline of 0.1833 ($z = +24.7$; Spearman $\rho = +0.240$, reproducing held-out and on the test split, and surviving a crack-size control at partial $\rho = +0.199$). The latent partially images itself into the visible state. The practical size of the leak is small — a best guess moves from 0.18 to 0.27 against a mean pairwise IoU of 0.283 — so the conditional remains broad, but we report this as a measured *floor* on achievable sharpness rather than claiming a clean hidden-variable setup.

The branching window is narrow. The stochasticity is concentrated in roughly three of nineteen stored steps. The earliest steps are frozen (per-step change below the cross-sample spread, so an autoregressive model wins by copying), a short window branches sharply (nearest neighbours diverge by factors of 3.1, 9.1 and 4.9 over consecutive steps), and thereafter the outcome is locked (top-1 nearest-neighbour retention returns to 40/40). A one-step score averaged over the trajectory therefore dilutes the real signal roughly six-fold, and must be reported step-resolved.

3.5 Two negative results on candidate slots

We record two datasets we prepared and then rejected, because both failures are general.

A chaotic ensemble is not an aleatory one. A radiative-mixing-layer ensemble with nine cooling-time values and ten seeds appears ideal. It is not, for the task an autoregressive simulator actually trains: the seeds differ *only* in an initial velocity perturbation, and density and pressure at the first step are bitwise identical across seeds. That perturbation is stored in the state rows the model reads as input. The seed is therefore fully observed, and $p(u_{t+1} | u_t, c)$ is a point mass for a deterministic solver on a fully resolved state. Genuine spread exists only for the trajectory-given-condition map, where cross-seed spread grows to exceed the cross-condition spread by the final step. The dataset is a legitimate *chaotic ensemble-spread* benchmark; it is not an aleatory-uncertainty benchmark, and one-step CRPS on it degenerates toward mean absolute error. Two replicates per condition in a standard 80/10/10 split — a 75.6% relative standard error on the spread estimate and a three-bin rank histogram — would in any case not support a calibration claim.

An unconditional generative task is not a UQ benchmark, and grain labels are not a field. A Potts kinetic Monte Carlo grain-growth set with one thousand seeds is a textbook stochastic process, but it carries no conditioning variable at all: every sample presents an identical input, so $p(y | x) \equiv p(y)$. Worse, the target is a labelling, not a field. Relabelling a sample by a permutation that provably preserves the partition (boundary-indicator IoU exactly 1.0) scores a mean squared error statistically indistinguishable from comparing two *genuinely different* microstructures, while a constant field scores twice as well as either. Under a pointwise loss, mode collapse is the global optimum. This survives the obvious repair: even on a permutation-invariant 0/1 grain-boundary target, a spatially constant predictor attains a Brier score of 0.2469 against the true ensemble’s 0.2624 — the constant *beats* the truth by 5.9% — and a spatially scrambled ensemble ties it, because the per-node marginal standard deviation (0.1568) sits at the independent-Bernoulli null (0.1572). Per-node scoring here is not weak but anti-informative. The generative-microstructure community’s choice of morphological-distribution metrics [13, 48] is correct, and a pointwise proper score is simply the wrong instrument.

3.6 The benchmark that should exist

We state the design requirements explicitly, each traceable to a failure above. A one-to-many CAE benchmark needs: (1) at least 50 members at each of at least 50 conditions, with fair-CRPS reported under the finite-ensemble correction [8]; (2) a native mesh with connectivity retained and nodal fields, since every dataset that regriddd destroyed exactly the structure a hierarchical mesh network exists to exploit; (3) a *labelled uncertainty-source axis*, because no CAE benchmark today distinguishes “my solver with a different seed” from “a different solver”; (4) a paired deterministic control on which the predicted spread must collapse, without which a model that hallucinates spread is indistinguishable from a calibrated one; and (5) an engineering quantity-of-interest distribution, since calibration on a stress field that is miscalibrated on peak stress is worthless.

4 Methodology

4.1 Problem statement: the conditional law, not the conditional mean

Let g denote everything a learned simulator observes about a scene: the mesh $(\mathcal{V}, \mathcal{E})$, the nodal coordinates, the boundary-condition and material channels, and any scalar conditions supplied alongside them. Let $\mathbf{y} \in \mathbb{R}^{N \times F}$ be the field to be predicted at the N mesh nodes. The deterministic simulator lineage—GNS, MeshGraphNets, and the hierarchical variants including HI-MGN—learns a map $\hat{\mathbf{y}} = f_\theta(g)$ under a pointwise regression loss,

$$\mathcal{L}_{\text{MSE}}(\theta) = \mathbb{E}_{(g, \mathbf{y})} \|f_\theta(g) - \mathbf{y}\|_2^2, \quad (1)$$

whose population minimiser is the conditional mean $f^*(g) = \mathbb{E}[\mathbf{y} \mid g]$.

The conditional mean is optimal for squared-error point prediction, but does not specify uncertainty when $p(\mathbf{y} \mid g)$ is broad. Write the data-generating process as

$$\mathbf{y} = \mathcal{S}(g, \boldsymbol{\xi}), \quad \boldsymbol{\xi} \sim p(\boldsymbol{\xi} \mid g), \quad (2)$$

where $\mathcal{S}$ is the physical solver and $\boldsymbol{\xi}$ is a latent variable that governs the response but is *not* part of g : an unmeasured geometric imperfection, a process-to-process manufacturing deviation, a microstructural realisation. The quantity of engineering interest is then the conditional law

$$p(\mathbf{y} \mid g) = \int p(\mathbf{y} \mid g, \boldsymbol{\xi}) p(\boldsymbol{\xi} \mid g) d\boldsymbol{\xi}, \quad (3)$$

and a model trained under (1) returns a single point of it. When (3) is multimodal—as it is for mode selection in buckling, or for warpage branches in injection moulding—the conditional mean need not lie in the support of $p(\mathbf{y} \mid g)$ at all: it is an average of physically realisable states that is itself not a physically realisable state. Tolerance analysis, yield prediction and reliability assessment all require the width and shape of (3), not its centroid.

We therefore ask a model to emit *samples* and compare the generated and reference ensembles. Section 5 distinguishes the historical SAOI scalar-distance metric from the implemented buckling energy score. The two mesh-native constructions below place stochasticity in a graph latent or directly in the field. This section describes the current implementation; the historical SAOI results in Section 6 are not a validation of every subsequently added option.

4.2 Shared backbone

Both models inherit the encode–process–decode structure of MeshGraphNets and the hierarchical processor of HI-MGN. Node and edge features are embedded by MLPs into latent vectors $\mathbf{v}_i$ and $\mathbf{e}_{ij}$, and each message passing (MP) block performs

$$\mathbf{e}'_{ij} = f_E(\mathbf{e}_{ij}, \mathbf{v}_i, \mathbf{v}_j), \quad (4)$$

$$\mathbf{v}'_i = f_V\left(\mathbf{v}_i, \sum_{j \in \mathcal{N}(i)} \mathbf{e}'_{ij}\right), \quad (5)$$

with residual connections, after which a decoder MLP reads the nodal output. The receptive field expands by at most one mesh hop per block; communicating between arbitrary nodes through this path alone requires depth at least the graph diameter. HI-MGN replaces the flat stack with a multiscale processor. Farthest-point sampling chooses seeds and multi-source breadth-first search assigns nodes to their nearest seed in graph distance. Coarse edges are induced by fine edges

crossing cluster boundaries. In `voronoi_seedmean` mode, features are mean-pooled while coarse positions use the seed nodes. MP blocks run at each level, and a learned graph interpolation network reconstructs fine-resolution features on the way back up. For example, the current SAOI arm-1 configurations specify two coarsening levels with 1000 and 100 requested coarse nodes, hidden width 128, and a 4, 6, 8, 6, 4 MP schedule. Other arms vary capacity or conditioning. Sharing this hierarchy does not make the full models identical: posterior, prior, input dimensions, modulation and training objectives also differ.

A note on the input contract. The `cond_var` channels following the state rows in `nodal_data` are *conditions*: quantities the model reads but never predicts (process settings, known boundary values). They enter $\mathbf{x}$ alongside the state, positional and node-type features, receive their own normalisation statistics, and are retained as known inputs during an autoregressive rollout. For the static $T = 1$ contract, the state input is zero and the target is the absolute output field; temporal data instead use the next-step increment as target. The conditions are part of g ; the latent ξ of (2) is, by construction, not.

4.3 MeshGraphNets-V: a latent-variable simulator

MeshGraphNets-V (MGN-V) uses a conditional variational-encoder architecture with an aggregate-posterior MMD objective, rather than a KL-based ELBO. A graph-level latent $\mathbf{z} \in \mathbb{R}^{K \times d}$ carries the sample-to-sample spread while the processor carries what geometry and boundary conditions determine. The latent is organised as $K = L + 1$ *slots*, one per hierarchy level plus the fine level, each of width d ; the current SAOI configuration uses $K = 3$, $d = 16$, i.e. 48 latent coordinates in total. These are graph-level slots injected at different levels, not a latent vector at every coarse node.

Posterior. A GNN encoder sees the normalised target and edge features and, when `vae_graph_aware=True` as in the current SAOI configurations, also the normalised input-node features. It runs message passing and pools to a single graph vector by attention, from which all slots are emitted:

$$q_\phi(\mathbf{z} \mid \mathbf{y}, g) = \mathcal{N}(\boldsymbol{\mu}_\phi(\mathbf{y}, g), \text{diag } \boldsymbol{\sigma}_\phi^2(\mathbf{y}, g)). \quad (6)$$

Decoder. The HI-MGN simulator itself is the decoder. Each hierarchy level receives its corresponding slot at every MP block. The implementation supports a linear concatenation fuser and AdaLN-Zero modulation; the current SAOI configurations select AdaLN-Zero. The posterior variance is diagonal conditional on its inputs, although mixing over targets can induce dependence between slots.

Conditional prior. At inference the target is unavailable. With the learned conditional prior enabled, $\mathbf{z}$ is drawn from $p_\psi(\mathbf{z} \mid g)$. Its default family is conditional flow matching in latent space: a velocity network transports $\mathcal{N}(\mathbf{0}, \mathbf{I})$ onto the posterior cloud for that graph, conditioned on a pooled graph embedding, and is integrated by a Heun or Euler ODE solver. All slots are flattened jointly, so cross-slot dependence is learned rather than assumed factorised. The implemented path retains endpoint noise with standard deviation 10^{-4} in the prior’s working coordinates. An optional frozen-prior training phase freezes the encoder and decoder, standardises posterior latents using training-set moments, and fits only the prior; sampling undoes that affine standardisation. A legacy graph-conditioned Gaussian mixture and standard-normal latent sampling are also available. They are distinct inference choices, not equivalent priors.

Objective. For joint training with the flow-matching prior, the current loss is

$$\mathcal{L}_{\text{MGN-V}} = \alpha_{\text{recon}} \mathcal{L}_{\text{recon}} + \lambda_{\text{MMD}} \mathcal{L}_{\text{MMD}} + \beta_{\text{aux}} \mathcal{L}_{\text{aux}} + w_{\text{prior}} \mathcal{L}_{\text{FM}}, \quad (7)$$

where $\mathcal{L}_{\text{recon}}$ is the channel-weighted, node-averaged posterior reconstruction error in normalised target coordinates (MSE in the current SAOI configurations; Huber is also supported). $\mathcal{L}_{\text{MMD}}$ averages a multi-bandwidth RBF MMD squared over slots, comparing minibatch posterior samples with independent $\mathcal{N}(0, I)$ samples. It uses the biased V-statistic, including diagonal kernel terms, with fixed or detached median-based bandwidths. This is aggregate-posterior regularisation in the InfoVAE sense [51]; it does not match each conditional posterior to the learned $p_{\psi}(\mathbf{z} \mid g)$, or constrain the joint cross-slot law. $\mathcal{L}_{\text{aux}}$ is the graph-averaged squared error between predicted and target peak-to-valley values of a selected normalised output channel. For $T = 1$ this is the field statistic; for temporal data it is the increment statistic. It encourages reconstruction of that statistic without guaranteeing calibrated spread or a mutual-information lower bound. $\mathcal{L}_{\text{FM}}$ is the flow-matching loss that fits the conditional prior to a *fresh* posterior sample drawn each step. A switch $g_{\text{enc}} \in [0, 1]$ scales the gradient that $\mathcal{L}_{\text{FM}}$ sends back through the encoder; at $g_{\text{enc}} = 0$ the prior chases a detached posterior, and at $g_{\text{enc}} = 1$ the encoder also receives the velocity-regression gradient through the path point and target. Section 4.5 shows why that second coupling is not the variational rate term it resembles.

4.4 cHI-MGNflow: flow matching in field space

cHI-MGNflow places the stochasticity at the output instead of in a bottleneck. It learns a conditional flow directly on the $N \times F$ field. Let $\mathbf{y}_1$ be the normalised target and $\mathbf{y}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ an independent Gaussian source drawn at every node and channel. With $s = 1 - \sigma_{\text{min}}$ the conditional path and its velocity are, with the implemented $\sigma_{\text{min}} = 10^{-4}$,

$$\mathbf{y}_t = (1 - st) \mathbf{y}_0 + t \mathbf{y}_1, \quad (8)$$

$$\mathbf{u} = \mathbf{y}_1 - s \mathbf{y}_0, \quad (9)$$

and the network regresses the velocity,

$$\hat{\mathcal{L}}_{\text{flow}} = \frac{\sum_b w(t_b) \sum_{i=1}^{N_b} \sum_f a_f (v_{\theta}(\mathbf{y}_{t_b}, t_b, g_b)_{if} - u_{bif})^2}{\sum_b N_b w(t_b)}, \quad (10)$$

where a_f are nonnegative channel weights normalised to sum to one, and b indexes graphs in the minibatch. The implementation normalises by the realised sum of node-time weights (with a numerical denominator floor), not by an expectation over t . Without time weighting, $w(t) = 1$. The flow time t is drawn *per graph*, not per node: the velocity field is a function of the whole field state, so mixing times inside one graph would ask the network to denoise a state lying on no path at all. The hierarchical processor is traversed once per velocity evaluation, with t entering through AdaLN modulation and the noisy field entering alongside the conditions.

The default `flow_head=v` predicts velocity directly. An optional data-prediction weighting, called `x0` in the configuration, uses $w(t) = (1 - st)^2$. The per-node loss identity is exact:

$$\hat{\mathbf{y}}_1 = s \mathbf{y}_t + (1 - st) \hat{\mathbf{u}} \implies \|\hat{\mathbf{y}}_1 - \mathbf{y}_1\|^2 = (1 - st)^2 \|\hat{\mathbf{u}} - \mathbf{u}\|^2, \quad (11)$$

The batch normalisation in (10) still applies. Separately, the current code supports `flow_head=x`, whose network output is a clean-field estimate h_{θ} and whose returned velocity is

$$v_{\theta} = \frac{h_{\theta} - s \mathbf{y}_t}{\max(1 - st, \varepsilon_{\text{head}})}. \quad (12)$$

The default floor is $\varepsilon_{\text{head}} = 0.05$. This conversion equals the unfloored path identity only where $1 - st \geq \varepsilon_{\text{head}}$. Without this floor, the gain reaches 10^4 at $t = 1$; it is finite at the implemented positive σ_{min} and becomes singular only in the zero-noise limit. Thus output-head choice and time-loss weighting are separate options.

Time sampling can be uniform or logit-normal, with an optional probability of pinning a graph to $t = 0$. At that endpoint, independence of $\mathbf{y}_0$ and $\mathbf{y}_1$ gives the population regression optimum $v^*(\mathbf{y}_0, 0, g) = \mathbb{E}[\mathbf{y}_1 | g] - s\mathbf{y}_0$. The implemented one-evaluation readout is $\mathbf{y}_0 + v_\theta(\mathbf{y}_0, 0, g)$, which at this optimum equals $\mathbb{E}[\mathbf{y}_1 | g] + \sigma_{\text{min}}\mathbf{y}_0$. Its error also depends on how well the trained network fits $t = 0$.

Sampling integrates the learned ODE from $t = 0$ to $t = 1$ with Euler or Heun, holding the graph hierarchy fixed. Even with the exact marginal velocity and exact integration, the stated path ends at $\mathbf{y}_1 + \sigma_{\text{min}}\mathbf{y}_0$, so its target law is the Gaussian-smoothed data law. A finite network and finite integration budget introduce additional approximation error.

4.5 Why a trainable-target flow objective shrinks the posterior

A tempting reading of (7) is that opening the encoder gradient ($g_{\text{enc}} = 1$) turns $\mathcal{L}_{\text{FM}}$ into the $D_{\text{KL}}(q(\mathbf{z} | \mathbf{y}, g) \| p(\mathbf{z} | g))$ rate term of a conditional VAE, making (7) an ELBO. It does not, and the difference has a direction.

The flow-matching theorem [25] equates the gradients of the conditional and marginal velocity-regression objectives *with respect to the velocity model’s parameters, for a fixed target distribution*. It says nothing about gradients with respect to parameters that *move* the target. A VAE rate term contains the posterior entropy and vanishes whenever $q = p$; a flow-matching MSE is not.

The consequence is quantitative. Consider the scalar Gaussian instance $Z_0 \sim \mathcal{N}(0, 1)$, $Z_1 \sim \mathcal{N}(0, a^2)$ independent, with the linear path $Z_t = (1 - t)Z_0 + tZ_1$ and velocity target $U = Z_1 - Z_0$. Here a plays the role of the trainable posterior scale, with $a > 0$. This example sets $\sigma_{\text{min}} = 0$ to give a simple closed form. Since (U, Z_t) is jointly Gaussian, the optimal regressor is the conditional expectation and its irreducible loss is the conditional variance,

$$\text{Var}(U | Z_t) = (1 + a^2) - \frac{(a^2t - (1 - t))^2}{(1 - t)^2 + a^2t^2} = \frac{a^2}{(1 - t)^2 + a^2t^2}. \quad (13)$$

Integrating over a uniform time schedule gives the loss floor attained by a *perfectly* trained velocity model,

$$\int_0^1 \text{Var}(U | Z_t) dt = \int_0^1 \frac{a^2}{(1 - t)^2 + a^2t^2} dt = \frac{\pi a}{2}. \quad (14)$$

The floor is *linear in a* . Meanwhile a prior that matches the target exactly has $D_{\text{KL}} = 0$ for every a . So the optimised flow-matching objective can always be decreased by shrinking the target distribution, even though the density-matching problem it is supposed to represent is already solved perfectly at every scale. In this Gaussian example, allowing the encoder scale to optimise the best-achievable FM loss therefore creates contraction pressure even with unlimited prior capacity. For the implemented positive endpoint noise, the corresponding variance is $a^2/[(1 - st)^2 + a^2t^2]$; its derivative with respect to a is $2a(1 - st)^2/[(1 - st)^2 + a^2t^2]^2 > 0$, so the scale incentive remains in this example. This calculation concerns the FM term alone, without reconstruction, MMD, variance floors or latent standardisation.

The counterexample shows that a trainable-target FM loss is not generally a VAE rate term. It does not establish the net effect of the coupling in a nonlinear, jointly trained model, nor rule out an empirical benefit. The historical SAOI note reports a small observed contrast for this factor

(Section 6); the incomplete, unreplicated sweep cannot prove that the causal effect is zero or identify the cause of under-dispersion.

5 Evaluation protocol

A distributional claim is only as good as the rule used to test it. Three properties of this setting make the obvious choices misleading, and each one has a concrete remedy that we adopt throughout.

5.1 Proper scores and the ensemble-size correction

Let $\{\mathbf{x}_i\}_{i=1}^m$ be the model’s ensemble for a scene and $\mathbf{y}$ an observation. The energy score

$$\text{ES}(\{\mathbf{x}_i\}, \mathbf{y}) = \frac{1}{m} \sum_{i=1}^m \|\mathbf{x}_i - \mathbf{y}\|_2 - \frac{1}{2m^2} \sum_{i=1}^m \sum_{j=1}^m \|\mathbf{x}_i - \mathbf{x}_j\|_2 \quad (15)$$

is strictly proper for the full multivariate law, and its univariate specialisation is the continuous ranked probability score (CRPS). The estimator in (15) is, however, biased upward at small m , and the bias depends on m : a model emitting 8 draws is penalised against one emitting 64 for reasons having nothing to do with the distribution it represents. We therefore use throughout the *fair* estimator, which normalises the repulsion term by $m(m-1)$ rather than m^2 ,

$$\widetilde{\text{ES}} = \frac{1}{m} \sum_i \|\mathbf{x}_i - \mathbf{y}\|_2 - \frac{1}{2m(m-1)} \sum_i \sum_j \|\mathbf{x}_i - \mathbf{x}_j\|_2. \quad (16)$$

The correction is not cosmetic. On i.i.d. standard normal data the plain CRPS estimator returns 0.698 at $m = 4$ and 0.568 at $m = 64$ against a true value of 0.564, while the fair estimator is flat in m to within sampling noise over the same range.

5.2 Two baselines, because a raw score is unreadable

An energy score has no absolute scale, and the achievable value is not zero: the reference ensemble is itself a finite sample. We therefore report every score against two anchors.

The *self floor* splits the reference ensemble in half and scores one half against the other, repeated over random splits. It is the value a perfect model attains, and reporting a raw score without it invites reading an irreducible sampling floor as model error. The *mean-only reference* scores the reference ensemble’s own mean, repeated m times—a zero-spread model. A skill score

$$\text{SS} = \frac{\text{ES}_{\text{ref}} - \text{ES}}{\text{ES}_{\text{ref}} - \text{ES}_{\text{floor}}} \quad (17)$$

then reads 1 at the sampling floor and 0 for a model no better than predicting the mean.

We validated the implementation on synthetic Gaussian data where the answer is known. The true distribution scores 1.4844 against a self floor of 1.4934, and beats a distribution that is too narrow (1.5946), too wide (1.7054), biased (2.1392) and mean-only (2.1008)—i.e. propriety holds numerically and the floor is attained by the truth, as it must be.

5.3 Dispersion is a separate question from location

A proper score is a single number, and a single number cannot separate an ensemble that is correctly centred but too narrow from one that is correctly dispersed but displaced. Since these are exactly the two failure modes observed in Section 6, and they call for opposite interventions, we report dispersion diagnostics alongside every score.

The *dispersion ratio* $\text{sd}(\text{gen})/\text{sd}(\text{ref})$ has target 1; below 1 the ensemble is too narrow. The *bias* $(\overline{\text{gen}} - \text{ref})/\text{sd}(\text{ref})$ has target 0; a value above 1 is a mean offset that no width correction can repair. The *rank (PIT) histogram* places each reference realisation among the model’s own ensemble: flat indicates calibration, U-shaped indicates under-dispersion, domed indicates over-dispersion. Dispersion ratio and PIT tail mass are independent views of the same defect and should agree; on synthetic reproductions of the evaluation design used here, a calibrated ensemble reports PIT KS 0.12 and tail mass 0.016, while one that is $2\times$ too narrow reports tail mass 0.256 against a dispersion ratio of 0.574.

5.4 Quotienting exact symmetries

Where the conditional law possesses an exact symmetry, scoring raw fields measures the model’s luck in guessing a group element rather than its distributional accuracy. In the buckling benchmark of Section 6.2 the law $p(\mathbf{y} \mid g)$ is exactly $SO(2)$ -symmetric by construction, so all scores there are computed on a rotation-invariant descriptor—the magnitude of the circumferential Fourier transform at each axial station—which quotients the symmetry while retaining the mode content. The symmetry then doubles as a free correctness check on any learned distribution.

5.5 What CRPS cannot do here

One negative result is worth recording, because it determined the primary metric. On an evaluation set with a *single* truth per condition, the accuracy term of CRPS dominates its spread term, so a mean-collapsed model scores well. Under that design CRPS ranks regressors, not generative models, and we do not use it as a primary metric. The evaluation in Section 6.1 instead pairs each condition with 125 realisations, which makes the reference a genuine estimate of $p(\mathbf{y} \mid g)$ and gives the dispersion ratio a meaningful target of 1.

6 Numerical results

Two studies are reported. The first is a head-to-head between MGN-V and cHI-MGNflow on an industrial warpage dataset, which establishes the ranking and —more importantly—isolates two orthogonal failure modes. The second responds to a limitation of that evaluation which no amount of further training can repair, by constructing a benchmark in which the withheld variable is known by construction.

6.1 Injection-moulded warpage (SAOI)

6.1.1 Data and task

The SAOI dataset comprises injection-moulded parts sharing mesh topology within a family but differing in outcome through process-to-process variability. The model observes the mesh, nodal coordinates and three input-only condition channels (wall thickness and part identifiers); it predicts

Table 2: SAOI head-to-head, mean over three held-out part families. Lower W_1/sd is better; the dispersion ratio targets 1. MGN-V arm 2 died in training, so its design is 7/8 (see Section 6.1.7).

model	best arm	W_1/sd	$ \Delta\text{mean} /\text{sd}$	dispersion ratio
MeshGraphNets-V	7 (ad g1 c0 r001)	0.410	0.218	0.475
MeshGraphNets-V	3 (cc g1 c0 r100)	0.434	0.176	0.526
cHI-MGNflow	1 (b16 tu k0 lr1)	0.896	0.724	0.475

the three displacement components. The latent ξ of (2)—the realisation-to-realisation process deviation—is not among the inputs, which is precisely what makes $p(\mathbf{y} | g)$ broad.

The quantity of interest is the *warpage spread*, the peak-to-valley $\max_i z_i - \min_i z_i$ over nodes at the final step. It is the scalar that governs whether a part passes tolerance, and all distributional metrics below are computed on it and normalised by the reference spread’s own standard deviation so that families are comparable.

6.1.2 Protocol

Both models use an identical two-level HI-MGN hierarchy (1000 and 100 coarse nodes, 4, 6, 8, 6, 4 MP schedule, hidden width 128, four cached hierarchy variants), are trained with AMP, EMA and gradient checkpointing under DDP, and differ only in the probabilistic construction of Section 4. The z -displacement channel carries a $10\times$ loss weight, since warpage is the engineering target. MGN-V uses $K = 3$ latent slots of width 16.

Each model was given an 8-arm 2^{4-1} resolution-IV fractional factorial at 1000 epochs, one arm per GPU. The factors were, for MGN-V, z -conditioning, g_{enc} , capacity and λ_{MMD} ; for cHI-MGNflow, batch size, flow-time sampling, capacity and learning rate. Inference draws 2000 samples per scene against three held-out part families, each paired with 125 reference realisations of that same part. Because the reference is 125 realisations of the *very geometry being predicted*, it estimates the true conditional law, and the dispersion ratio has a meaningful target of 1.

6.1.3 Ranking

Table 2 and Figure 1(a) give the result. MGN-V wins by $2.2\times$ on the primary metric, and the two models do not overlap: cHI-MGNflow’s *worst* arm (1.604) is more than $3\times$ MGN-V’s worst (0.510). The separation is far larger than the arm-to-arm spread within either model, so it does not depend on hyperparameter selection.

6.1.4 The two models fail in orthogonal ways

The ranking is the less interesting result. Decomposing the error into location and width (Figure 2) shows two distinct defects.

cHI-MGNflow is competitive on two of the three families and collapses on the third, where W_1/sd runs 1.4–3.7. On that family the error is almost entirely a mean offset: arm 1 reports $W_1 = 1.425$ against $\Delta\text{mean} = 1.408$, and arm 7 reports 3.659 against 3.641. This is not a calibration problem but a failure to generalise the conditional mean to that geometry. Decisively, the *bias changes sign between families* (−0.13 to −0.44 on one, +1.4 to +3.6 on another): a unit or scaling bug would be one-directional, so the per-geometry conditional itself is wrong.

MGN-V on the same family reports Δmean between −0.21 and −0.02 with W_1 between 0.22 and 0.30. Its location is right; its distribution is too narrow.

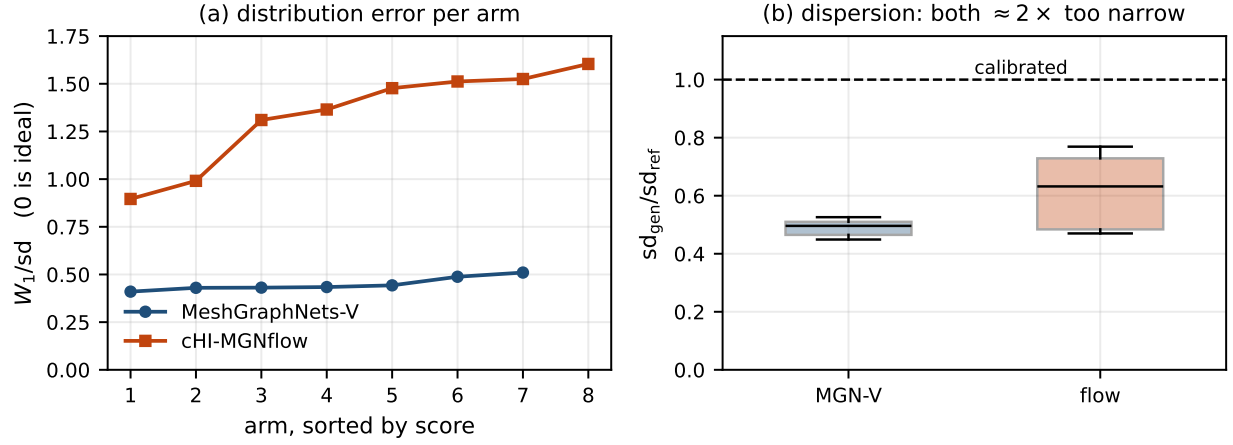


Figure 1: SAOI sweep. (a) Per-arm distribution error, arms sorted by score; the two models do not overlap. (b) Dispersion ratio; the target is 1 and neither model approaches it.

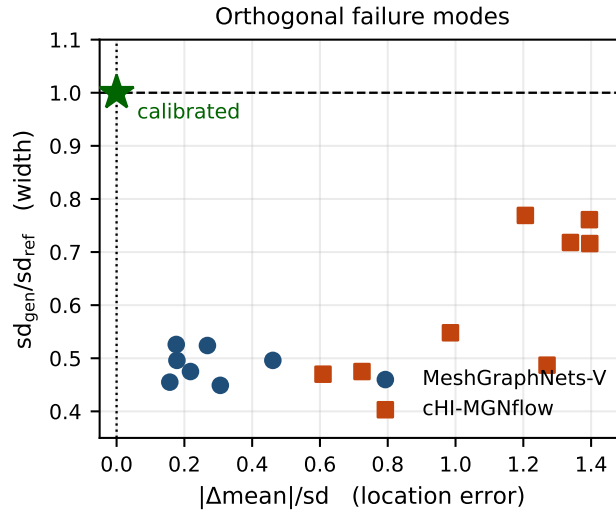


Figure 2: Location error against width for every arm of both models. The two methods occupy disjoint regions: MGN-V is correctly centred and too narrow, cHI-MGNflow spans a wide range of location error. The two defects call for opposite interventions, which is why a single proper score is not a sufficient reason.

6.1.5 Neither model is calibrated

This is the blocking result, and it is not visible in the ranking. The dispersion ratio should be 1; the best value achieved by either model is 0.769 and the best MGN-V value is 0.526 (Figure 1(b)). Per family it is worse: cHI-MGNflow reports 0.199–0.378 on one set, i.e. the generated ensemble spans 20–38% of the truth’s width.

Both models produce ensembles roughly $2\times$ too narrow. The stated goal—matching the distribution—is met by neither; MGN-V is merely less wrong. Under-dispersion of this size is disqualifying for the engineering use case, because a tolerance analysis that halves the predicted spread systematically under-predicts failure rates.

6.1.6 Main effects, and two falsified predictions

For cHI-MGNflow the factor effects are large relative to its 0.708 arm-to-arm range and the best predicted direction coincides with the best observed arm: uniform flow-time sampling beats logit-normal by $\Delta = 0.333$, batch 16 beats batch 32 by 0.288, learning rate contributes 0.083 and capacity is negligible. For MGN-V every effect is tiny—the largest, capacity, is $\Delta = 0.054$ against a 0.10 arm-to-arm range—and is treated as provisional.

Two predictions argued from mechanism were falsified, and both are recorded because they are the kind of reasoning this field publishes without testing.

First, g_{enc} —differentiating the flow-matching prior loss through the encoder—has *no measurable effect*: $\Delta = 0.001$ on W_1/sd , with dispersion ratio 0.482 at $g_{\text{enc}} = 0$ versus 0.494 at $g_{\text{enc}} = 1$, i.e. marginally *wider* if anything. The prediction had been that opening this gradient would let the encoder make $\mathbf{z}$ trivially predictable from the graph and so narrow the conditional distribution. Section 4.5 shows that the contraction pressure is real as a mechanism; the measurement shows its net effect at this budget is nil. The mechanism therefore rules the coupling out as a *fix* without explaining the observed deficit.

Second, logit-normal flow-time sampling was predicted to beat uniform, on the rationale that concentrating the budget near $t \approx 0.5$ spends it where the velocity is hardest. The opposite holds by cHI-MGNflow’s largest margin. On mesh fields the path endpoints evidently matter more than they do on images, and an image-derived default should not be transplanted here.

6.1.7 What this evaluation cannot say

Three limits bound every claim above. (i) MGN-V arm 2 was lost to a GPU fault, so the design is 7/8 and resolution IV does not strictly hold; the missing arm sits in the losing capacity group, so that conclusion in particular could flip. (ii) 1000 epochs is a budget, not convergence. (iii) Most importantly, *each eval set contains a single geometry*. The sweep therefore measures each model’s conditional law for one part per family, not across a population. The $2.2\times$ ranking and the under-dispersion hold for those three parts; nothing in this experiment says how either model behaves on a fourth, and nothing in it can separate a genuine failure to transfer a conditional distribution from a failure on three particular parts.

That limitation is structural rather than an oversight in execution: an industrial dataset with many realisations of many geometries, where the withheld variable is known, does not exist. That is what motivates Section 6.2.

6.2 A benchmark with a known withheld variable

Figure 3 summarises the construction: the mechanical setup, the two-component imperfection law that supplies the randomness, what a stored sample does and does not contain, and the design space. The remainder of this section states each of those in turn and then reports what was measured.

6.2.1 Design rationale

The evaluation above needs a dataset with three properties simultaneously: many geometries; many realisations per geometry; and a withheld latent whose law is *known*, so that the reference conditional distribution is a controlled object rather than an artefact of whatever the factory happened to do. Axially compressed thin cylindrical shells supply all three. Their post-buckling response is famously imperfection-sensitive: near the critical load a family of circumferential modes is nearly degenerate, and which one is selected is decided by geometric imperfections far too small to measure.

6.2.2 Construction

Shells are meshed analytically with S8R elements in double precision, with the element count set so that the buckling half-wavelength $\lambda_{1/2} = 1.728\sqrt{Rt}$ is resolved by a fixed number of elements at every geometry; across the box this holds 6.0 nodes per half-wave from 5 311 to 15 190 nodes. The imperfection has two components.

Component A is the deterministic process signature: ovalization plus a weld pattern, at RMS $0.15t$. It is identical for every draw of a geometry and is *given* to the model. Any harmonic falling within three wavenumbers of the critical one, $n_{\text{cr}} \approx 0.86\sqrt{R/t}$, is removed—without that filter the signature does not merely break symmetry but selects the outcome, which we verified by direct dose-response (mode $n = 8$ occurred 0/12 times with the component off and 12/12 times at $0.10t$ with the colliding harmonic present).

Component B is the withheld variable: a band-limited Matérn random field with RMS $\hat{\sigma} = 10^{-3}t$, correlation length $0.70\sqrt{Rt}$ and $\nu = 1.5$, stored as spectral coefficients so that a realisation transfers to a different mesh exactly rather than by interpolation. It is never written into the model’s inputs.

The ratio is the point: **Component B is $150\times$ smaller in RMS than the signature the model *can* see, and yet it determines the outcome completely.** A disturbance of 0.1% of the wall thickness selects which post-buckled branch the shell reaches. Because the same law is applied at every geometry, a model that learns the law rather than memorising the training modes should transfer.

Each draw is continued past the limit point by static loading to 70% of the target end-shortening, implicit dynamics through the snap, and a damped settle; convergence is measured per draw as the fractional change in field RMS over the last third of the settle window.

6.2.3 Dataset composition

Table 3 gives the composition: 22 geometries and 896 draws. The fields are packed in the shared mesh HDF5 contract with reference coordinates, the three displacement outputs, the Component A offset, nodal thickness and node type; the withheld Component B is stored in a separate group so that diagnostics can condition on it while a model reading the documented rows cannot.

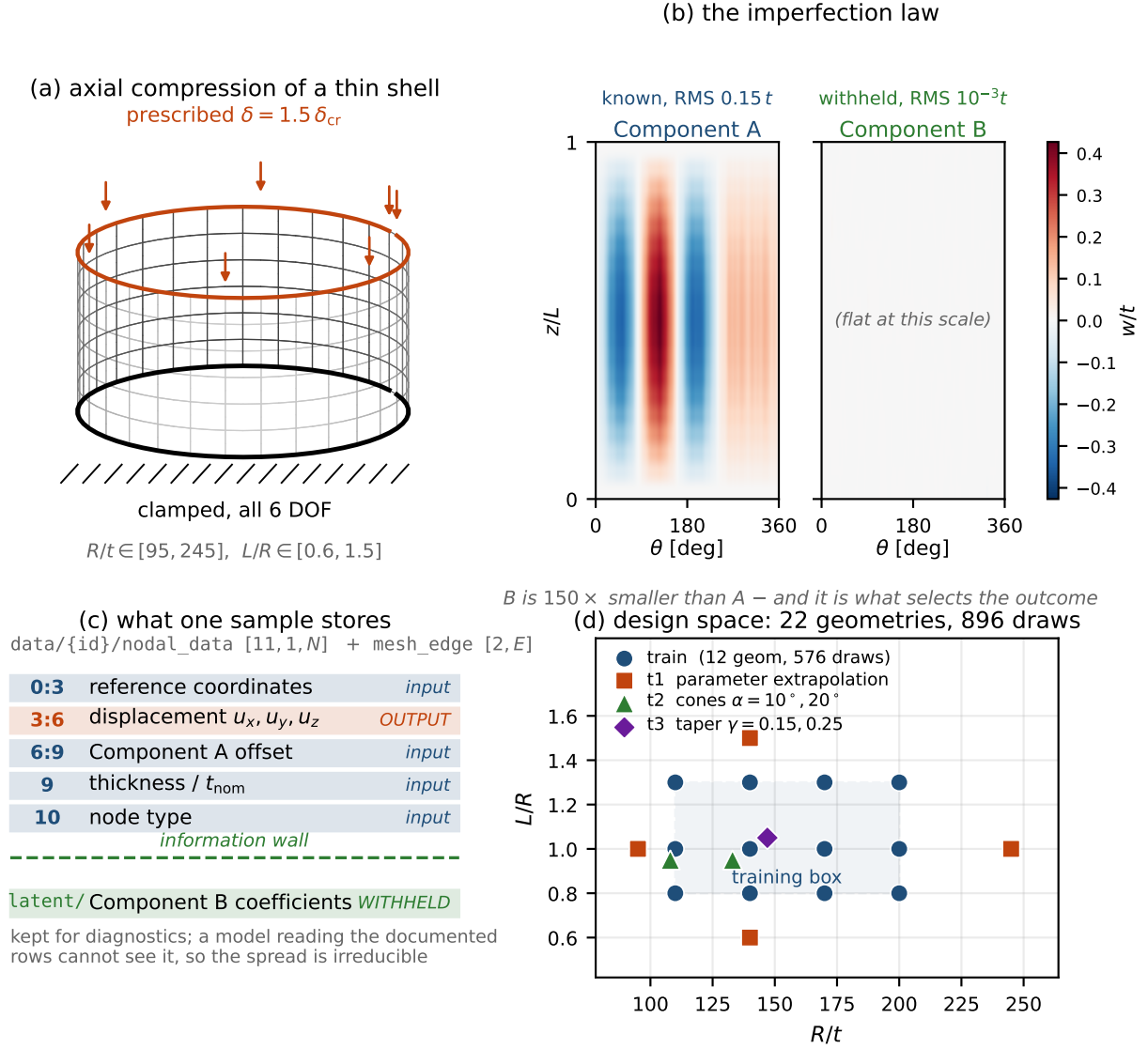


Figure 3: The benchmark. **(a)** Displacement-controlled axial compression of a thin shell, clamped at the base, driven to $1.5 \delta_{cr}$; the wireframe is the real generator mesh, drawn coarse for legibility. **(b)** The imperfection law, both components on *one* colour scale. Component A—ovalization plus a weld pattern, dominated by $n = 2$ —is deterministic per geometry and is given to the model. Component B is the withheld variable and is invisible at A’s scale, being $150 \times$ smaller in RMS, yet it alone decides which post-buckled branch the shell reaches. **(c)** The storage contract. Ten of the eleven rows are model inputs and three are the prediction target; the withheld coefficients live outside `nodal_data` entirely. **(d)** The design space, plotted from the plan file the production run consumed. Markers that would coincide (the cone and taper families sit at training-box parameters by construction) are offset slightly for legibility.

Table 3: Benchmark composition. Tiers t1–t3 are inference-only. Tiers t2 and t3 keep the *local* R/t inside the training range at every axial station, so a failure there is attributable to the structural novelty rather than confounded with parameter extrapolation, which t1 isolates separately.

tier	family	geometries	draws	local R/t	tests
train	cylinders	12	576	110–200	—
t1	cylinders	4	128	95, 245	parameter extrapolation
t2	cones, $10^\circ/20^\circ$	4	128	115–191	axially varying n_{cr}
t3	thickness taper	2	64	112–140	axially varying n_{cr}

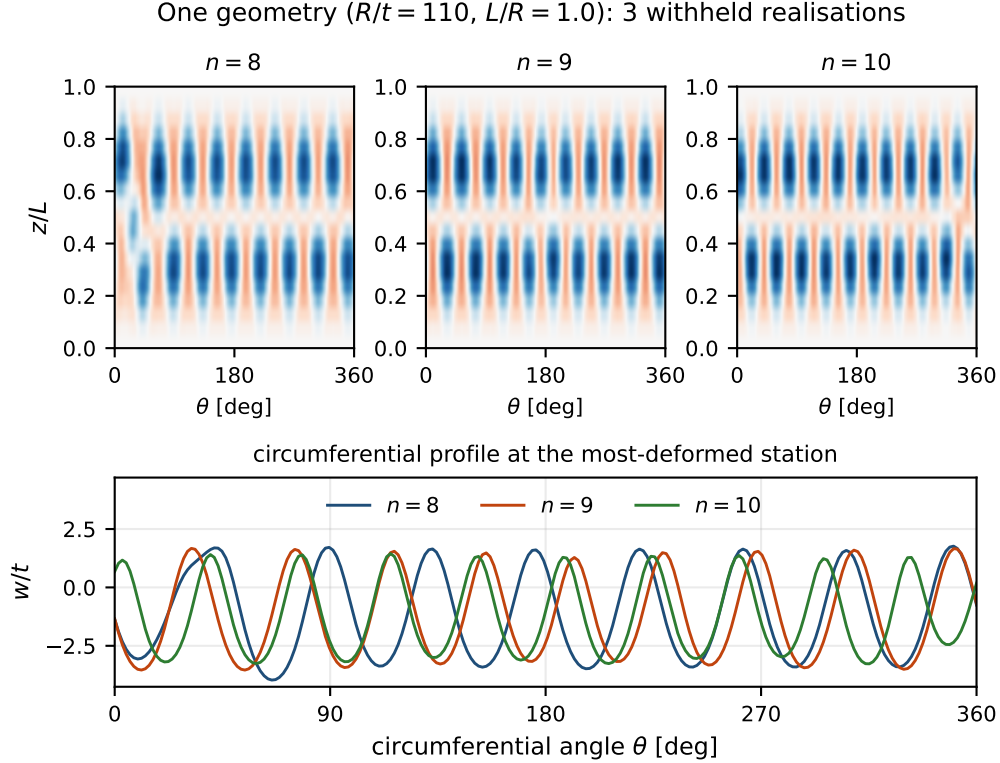


Figure 4: Three draws from one geometry, differing only in the withheld Component B realisation. Top: radial displacement w/t over the unrolled mid-surface. Bottom: circumferential profile at the most-deformed axial station, where the different wavenumbers are directly countable.

6.2.4 Characterisation

The training tier is complete: 12 geometries at 48 draws each, 576 draws, with *zero* solver failures out of 588 attempted including the tier that follows.

Figure 4 shows three realisations of one geometry. They are the same shell under the same load with the same visible inputs, and they buckle into different circumferential modes; the profile panel makes the wavenumbers directly countable. This is the one-to-many property the benchmark exists to supply.

Figure 5 reports the mode distributions. Every geometry supports three to five distinct modes, with a top-mode share between 38% and 67%, so no geometry is close to deterministic. Two trends

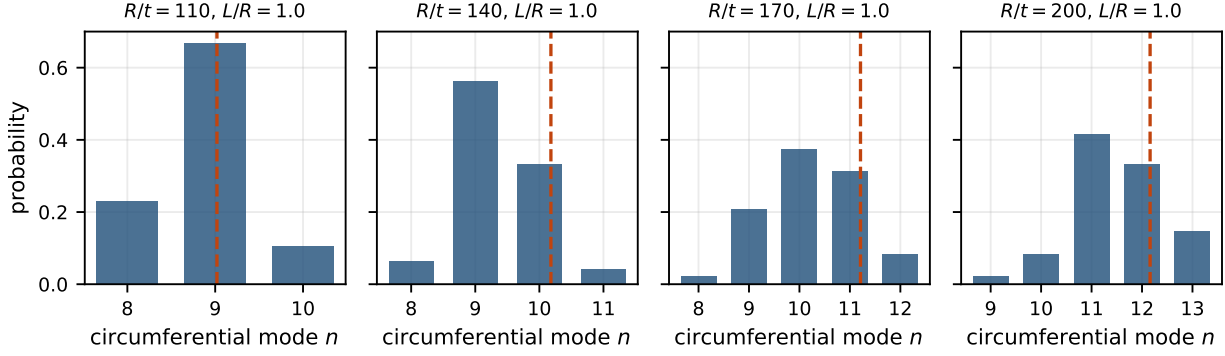


Figure 5: Measured mode distributions at fixed $L/R = 1.0$, 48 draws per geometry. The dashed line marks the classical prediction $n_{\text{cr}} = 0.86\sqrt{R/t}$. The distribution both *shifts* with R/t and *widens*, and the classical value drifts progressively to the right of the realised mode.

are visible across the tier and both are physically expected.

The distribution widens as the shell thins. The top-mode share falls from 65–67% at $R/t = 110$ to 38–56% at $R/t = 200$, with a correlation of -0.68 against R/t . Thinner shells have a denser cluster of near-degenerate modes near the critical load, so the withheld field has more nearly equivalent branches to choose between. The benchmark therefore contains a built-in difficulty axis: the conditional law is broader exactly where the shell is thinner.

The classical wavenumber over-predicts, systematically. The mean realised mode lies below $0.86\sqrt{R/t}$ at *every one* of the twelve geometries, by 1.6% to 13.0%, and the deficit grows with R/t (correlation $+0.59$ on its magnitude). The direction is the expected one—the closed form is derived for a perfect shell, whereas these are imperfect and buckle early into a marginally longer wavelength—but the size of the deviation at the thin end means the classical value should be treated as a reference line, not as a label the data is expected to reproduce.

Convergence is not uniform across the box, and this is a limitation. The settle window was calibrated on $R/t = 110$, $L/R = 1.0$ and does not transfer to the long shells. Grouping the retention-threshold exceedance by aspect ratio makes the pattern unambiguous:

L/R	median drift	draws over threshold
0.8	0.021–0.034	0% at every R/t
1.0	0.020–0.116	0–21%
1.3	0.038–0.191	0–81%

The driver is L/R , not R/t : the correlation of exceedance with R/t is only 0.28, while every $L/R = 0.8$ geometry converges completely and two of the three $L/R = 1.3$ geometries at higher R/t exceed the threshold on more than half their draws. An earlier frequency extraction had found the fundamental frequency of the *undeformed* shell nearly constant across the box (spread 1.014 $\times$), which is what motivated a single fixed settle window; that measurement does not govern the *post-buckled* relaxation timescale, and the present data shows the two are not interchangeable. The $L/R = 1.3$ rows should therefore be regarded as under-converged at the current settle length, their

drift value is stored per draw so the threshold can be re-applied, and the settle window needs to scale with L/R in any regeneration. The relationship between drift and near-degeneracy persists across the full tier (correlation -0.54 with the top-mode share), so the draws that fail to settle are preferentially the marginal ones—which is why they are retained and flagged rather than silently dropped.

6.2.5 The conditional law does depend on geometry

A distributional benchmark is worthless if the conditional distribution is in fact the same everywhere, so this was tested rather than assumed. On the discrete mode label, only 3 of 6 geometry pairs separate beyond the within-geometry sampling noise (total-variation distance 0.143 mean, 0.292 at the 95th percentile over 200 random half-splits). The three that fail are exactly those varying L/R at fixed R/t —which is correct, since n_{cr} depends on R/t alone.

On the field, all 12 ordered pairs separate. Scoring geometry A’s ensemble against geometry B’s reference and normalising by B’s own self floor gives off-diagonal values from 1.08 to 2.59 (mean 1.72), against a diagonal of ≈ 1 . The pairs that are invisible to the mode label separate at 1.08–1.28.

The conclusion matters beyond this dataset: **the argmax mode label is a lossy target**, and the field carries strictly more conditioning information than any scalar summary of it. This is the measured justification for scoring the invariant field descriptor rather than a discrete label—and it echoes the SAOI finding, where the entire evaluation rests on a single peak-to-valley scalar.

6.2.6 Status

The benchmark is a contribution of construction and characterisation. No model has yet been trained on it: generation is in progress, and the t2 and t3 families—cones and thickness tapers—have not been run at all, so whether they yield usable distributions is not yet established. We report it here because the SAOI evaluation cannot be repaired by better training, and because the properties that make a distributional benchmark valid are testable in advance of any model, as Figures 4 and 5 demonstrate.

7 Conclusions

We set out to move hierarchical mesh simulators from estimating a conditional mean to representing a conditional law, and compared two mesh-native ways of doing it: a latent-variable construction (MeshGraphNets-V) and a field-space flow (cHI-MGNflow). Both are built on the same HI-MGN backbone, so the comparison isolates where the stochasticity is placed.

The ranking, and why it is the least useful finding. On industrial warpage data MGN-V outperforms cHI-MGNflow by $2.2\times$ on normalised 1-Wasserstein distance, and the two do not overlap across an 8-arm design each. But the decomposition into location and width is what carries information: MGN-V is correctly centred and roughly $2\times$ too narrow, while cHI-MGNflow’s failure on one part family is almost pure mean offset, with a bias that *changes sign* between families—a per-geometry conditional that is wrong, not a calibration defect. These two failures call for opposite interventions, and neither is visible in a single proper score. Reporting dispersion and location alongside the score is therefore not optional bookkeeping.

The blocking result. Neither model is calibrated. The best dispersion ratio attained by either is 0.769, the best MGN-V value is 0.526, and per family cHI-MGNflow reaches 0.199. Ensembles

roughly half the correct width are disqualifying for the engineering purpose, since a tolerance analysis built on them under-predicts failure rates systematically. On this task the ranking question is settled and the calibration question is not, and the second is the one that matters.

A mechanism, stated precisely and then not over-claimed. Fitting a flow-matching prior to a jointly trained encoder is often described as supplying the rate term of a variational objective. It does not. We showed analytically that for the Gaussian instance the optimal conditional flow-matching loss floor is $\pi a/2$, *linear* in the trainable target scale a , while a prior matching the target exactly has zero KL at every a . The objective can therefore always be reduced by shrinking the target, and the identity gradient descent is following is not the one the ELBO analogy suggests. This applies to any method coupling a flow-matching prior to a trained encoder. We also measured its effect on this task and found it nil ($\Delta = 0.001$), so the mechanism rules the coupling out as a *fix* without accounting for the observed deficit—a distinction we state rather than elide.

Two predictions that did not survive. Opening the prior-to-encoder gradient was predicted to narrow the conditional distribution; its net effect is nil. Logit-normal flow-time sampling, the image-generation default, was predicted to beat uniform; uniform wins by cHI-MGNflow’s largest margin. On mesh fields the path endpoints matter more than they do on images.

The measurement gap, and a benchmark for it. Every claim above is bounded by the same limitation: each evaluation set holds a single geometry with 125 realisations, so the experiment measures each model’s conditional law for three particular parts and cannot separate a failure to *transfer* a conditional distribution from a failure on those parts. No industrial dataset resolves this, because the withheld variable is never known. We therefore constructed one: axially compressed shells in which the withheld imperfection is a specified Matérn field with RMS $10^{-3}t$ — $150\times$ smaller than the signature the model can see, and yet fully determining which post-buckled branch is reached. Across 22 geometries and four tiers, the out-of-distribution families are designed so that a failure is attributable: parameter extrapolation is isolated in one tier, while the cone and taper tiers keep local R/t inside the training range and vary only the structure.

Characterisation of the completed training tier shows the properties a distributional benchmark requires, and shows them *before* any model is trained: three to five modes per geometry with top-mode share 38–67%, a conditional law that broadens systematically as the shell thins (correlation -0.68 of top-mode share with R/t , so the benchmark carries a built-in difficulty axis), and—critically—a conditional law that genuinely varies with geometry, with all 12 ordered geometry pairs separating at field level. That last test also produced a result that reaches back into the SAOI evaluation: the discrete mode label separates only 3 of 6 pairs, so *the argmax label is a lossy target*. Any evaluation resting on a single scalar summary of the field—as the warpage peak-to-valley does—is measuring less than the data contains.

Limitations. The SAOI sweep lost one arm to a GPU fault, leaving a 7/8 design, and ran at a fixed 1000-epoch budget rather than to convergence; its factor effects for MGN-V are smaller than the arm-to-arm spread and should be treated as provisional. The benchmark is a contribution of construction and characterisation only: generation is still in progress, no model has been trained on it, and the cone and taper tiers have not been run, so whether they yield usable distributions remains open. One defect in it is already measured and should be stated plainly: the settle window was calibrated on a short shell and does not transfer, so the $L/R = 1.3$ geometries are under-converged, with up to 81% of their draws exceeding the retention threshold. The per-draw drift

is stored so the threshold can be re-applied, but that tier needs regeneration with an aspect-ratio-dependent settle length before it carries the same weight as the rest. We report the benchmark despite this because the measurement gap it addresses cannot be closed by training either model better.

Outlook. The two methods fail in complementary ways, which suggests the productive direction is not to pick one. cHI-MGNflow’s better native dispersion is worth borrowing; MGN-V’s correct conditional location is worth keeping. A coarse spatial generative model with conditionally sampled fine-scale refinement would place the stochasticity at the scale where the physical variability actually lives, rather than in a global bottleneck or at every node independently. Whether that closes the factor-of-two width deficit is an empirical question, and the benchmark introduced here is built to answer it.

A Benchmark generation cost

Every draw is an independent nonlinear continuation—static pre-buckling, implicit dynamics through the snap, damped settle—so generation cost is the practical constraint on how large a distributional benchmark can be. Because that constraint is rarely reported, we give the measurement.

Table 4 lists the mean wall-clock time per draw over the completed training tier, at 19 concurrent single-threaded solves. Cost is organised by the Batdorf parameter $Z = (L^2/Rt)\sqrt{1-\nu^2}$, the natural similarity group for cylindrical shells.

Table 4: Measured generation cost, 48 draws per geometry.

geometry	Z	nodes	mean [s]	range [s]
$R/t\ 110, L/R\ 0.8$	67	5 311	1 043	1 023–1 054
$R/t\ 140, L/R\ 0.8$	85	6 450	1 319	1 290–1 342
$R/t\ 170, L/R\ 0.8$	104	7 952	1 791	1 757–1 816
$R/t\ 110, L/R\ 1.0$	105	6 328	1 362	1 353–1 385
$R/t\ 200, L/R\ 0.8$	122	9 610	2 269	2 230–2 301
$R/t\ 140, L/R\ 1.0$	134	8 385	2 028	1 958–2 115
$R/t\ 170, L/R\ 1.0$	162	10 082	2 516	2 489–2 542
$R/t\ 110, L/R\ 1.3$	177	8 362	2 079	2 064–2 097
$R/t\ 200, L/R\ 1.0$	191	11 935	3 078	3 020–3 112
$R/t\ 140, L/R\ 1.3$	226	10 707	2 924	2 884–3 021
$R/t\ 170, L/R\ 1.3$	274	12 638	3 560	3 531–3 606
$R/t\ 200, L/R\ 1.3$	322	15 190	4 607	4 524–4 728

A power-law fit over the twelve geometries gives $\text{cost} \propto Z^{0.88}$. The within-geometry spread is remarkably tight—typically under 3%—which is worth noting in itself: although the *outcome* of a draw is stochastic, its *cost* is not, so the schedule is predictable even though the physics is not.

We record one planning error, because it is the kind that recurs. The schedule was estimated before launch from two measured points, which gave an exponent of 0.60 and a projection of 454 core-hours. The twelve-point fit gives 0.88 and 541 core-hours, so the two-point estimate under-predicted by $1.19\times$. A two-point calibration of a power law has no residual and therefore reports no uncertainty at all; that it happened to be optimistic here is luck, but that it carried no error bar is structural.

Mesh resolution is held fixed in the physically meaningful sense rather than in node count: the element size is set so the buckling half-wavelength $\lambda_{1/2} = 1.728\sqrt{Rt}$ is resolved by a constant number of elements at every geometry, which holds at 6.0 nodes per half-wave from 5 311 to 15 190 nodes. The larger geometries are therefore genuinely resolved, not merely more expensive.

B Model and training configuration

Both models in Section 6.1 share the backbone configuration in Table 5; the entries below the rule are what distinguishes them. Sharing everything above the rule is what makes the comparison a test of where the stochasticity is placed rather than of two separately tuned systems.

Table 5: Configuration. Values above the rule are identical for both models.

<i>shared</i>		
hidden width	128	
hierarchy	2 levels, 1000 and 100 coarse nodes	
coarsening	farthest-point sampling + Voronoi	
MP schedule	4, 6, 8, 6, 4 (down arm, coarsest, up arm)	
hierarchy variants	4 independently seeded, cached	
edge features	8 (deformed and reference dx, dy, dz , distance)	
positional features	4, rotation-invariant	
inputs / outputs	3 displacement channels, 3 condition channels	
loss weights	0.1, 0.1, 1 (z is the warpage target)	
input noise	$\sigma = 0.01$, training only	
augmentation	random z -rotation and reflection	
precision	bfloat16 AMP, EMA 0.99, gradient checkpointing	
split	80/10/10, seed 42	
epochs	1000 (a budget, not convergence)	
	MeshGraphNets-V	cHI-MGNflow
stochasticity	$K = 3$ latent slots, width 16	field-space flow
prior	conditional flow matching	—
integrator	Heun	Euler / Heun
swept factors	z -conditioning, g_{enc} , capacity, λ_{MMD}	batch size, t sampling, capacity, learning rate

Each sweep is a 2^{4-1} resolution-IV fractional factorial, one arm per GPU. Inference draws 2000 samples per scene. One caveat on the cHI-MGNflow design: batch size was varied at fixed epoch count, which also varies the number of optimiser updates, so the reported advantage of the smaller batch is partly an update-budget effect and should not be read as a pure batch-size result.

References

- [1] Michael S. Albergo and Eric Vanden-Eijnden. Building normalizing flows with stochastic interpolants. In *International Conference on Learning Representations (ICLR)*, 2023.
- [2] Michael S. Albergo, Nicholas M. Boffi, and Eric Vanden-Eijnden. Stochastic interpolants: A unifying framework for flows and diffusions, 2023.

- [3] Mohammad Babaeizadeh, Chelsea Finn, Dumitru Erhan, Roy H. Campbell, and Sergey Levine. Stochastic variational video prediction. In *International Conference on Learning Representations (ICLR)*, 2018.
- [4] Christopher Bülte, Philipp Scholl, and Gitta Kutyniok. Probabilistic neural operators for functional uncertainty quantification. *Transactions on Machine Learning Research*, 2025. ISSN 2835-8856. URL <https://openreview.net/forum?id=gangoPXSRw>.
- [5] Yadi Cao, Menglei Chai, Minchen Li, and Chenfanfu Jiang. Efficient learning of mesh-based physical simulation with bi-stride multi-scale graph neural network. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *Proceedings of Machine Learning Research*, pages 3541–3558. PMLR, 2023.
- [6] Emily Denton and Rob Fergus. Stochastic video generation with a learned prior. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, volume 80 of *Proceedings of Machine Learning Research*, pages 1174–1183, 2018.
- [7] Mohamed Elrefaie, Florin Morar, Angela Dai, and Faez Ahmed. DrivAerNet++: A large-scale multimodal car dataset with computational fluid dynamics simulations and deep learning benchmarks. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024) Track on Datasets and Benchmarks*, 2024.
- [8] Christopher A. T. Ferro. Fair scores for ensemble forecasts. *Quarterly Journal of the Royal Meteorological Society*, 140(683):1917–1923, 2014. doi: 10.1002/qj.2270.
- [9] Meire Fortunato, Tobias Pfaff, Peter Wirsberger, Alexander Pritzel, and Peter Battaglia. Multiscale meshgraphnets. In *2nd AI4Science Workshop at the 39th International Conference on Machine Learning (ICML)*, 2022.
- [10] Cristina Garcia Cardona, Gregory J. Wagner, Veena Tikare, Elizabeth A. Holm, Steven J. Plimpton, Aidan P. Thompson, Alexander Slepoy, Xiao W. Zhou, Corbett C. Battaille, and Michael E. Chandross. Crossing the mesoscale no-man’s land via parallel kinetic Monte Carlo. Technical Report SAND2009-6226, Sandia National Laboratories, 2009.
- [11] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.
- [12] Thomas M. Hamill. Interpretation of rank histograms for verifying ensemble forecasts. *Monthly Weather Review*, 129(3):550–560, 2001. doi: 10.1175/1520-0493(2001)129<0550:IORHFV>2.0.CO;2.
- [13] Nathan Hoffman, Cashen Diniz, Dehao Liu, Theron Rodgers, Anh Tran, and Mark Fuge. Grainpaint: A multi-scale diffusion-based generative model for microstructure reconstruction of large-scale objects. *Acta Materialia*, 288:120784, 2025. doi: 10.1016/j.actamat.2025.120784.
- [14] Eyke Hüllermeier and Willem Waegeman. Aleatoric and epistemic uncertainty in machine learning: an introduction to concepts and methods. *Machine Learning*, 110(3):457–506, 2021. doi: 10.1007/s10994-021-05946-3.
- [15] Steve Kench and Samuel J. Cooper. Generating three-dimensional structures from a two-dimensional slice with generative adversarial network-based dimensionality expansion. *Nature Machine Intelligence*, 3:299–305, 2021. doi: 10.1038/s42256-021-00322-1.

- [16] Georg Kohl, Li-Wei Chen, and Nils Thuerey. Benchmarking autoregressive conditional diffusion models for turbulent flow simulation. *Neural Networks*, 199:108641, 2026. doi: 10.1016/j.neunet.2026.108641.
- [17] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems 30 (NIPS 2017)*, 2017.
- [18] Simon Lang, Mihai Alexe, Mariana C. A. Clare, Christopher Roberts, Rilwan Adewoyin, Zied Ben Bouallègue, Matthew Chantry, Jesper Dramsch, Peter D. Dueben, Sara Hahner, Pedro Maciel, Ana Prieto-Nemesio, Cathal O’Brien, Florian Pinault, Jan Polster, Baudouin Raoult, Steffen Tietsche, and Martin Leutbecher. AIFS-CRPS: ensemble forecasting using a model trained with a loss function based on the continuous ranked probability score. *npj Artificial Intelligence*, 2:18, 2026. doi: 10.1038/s44387-026-00073-7.
- [19] Alex X. Lee, Richard Zhang, Frederik Ebert, Pieter Abbeel, Chelsea Finn, and Sergey Levine. Stochastic adversarial video prediction, 2018.
- [20] Zijie Li, Anthony Zhou, and Amir Barati Farimani. Generative latent neural PDE solver using flow matching, 2025. URL <https://arxiv.org/abs/2503.22600>.
- [21] Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations (ICLR)*, 2021. URL <https://openreview.net/forum?id=c8P9NQVtmn0>.
- [22] Zongyi Li, Nikola Borislavov Kovachki, Chris Choy, Boyi Li, Jean Kossaifi, Shourya Prakash Otta, Mohammad Amin Nabian, Maximilian Stadler, Christian Hundt, Kamyar Azizzadenesheli, and Anima Anandkumar. Geometry-informed neural operator for large-scale 3D PDEs. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023.
- [23] Mario Lino and Nils Thuerey. One scale at a time: Scale-autoregressive modeling for fluid flow distributions, 2026.
- [24] Mario Lino, Tobias Pfaff, and Nils Thuerey. Learning distributions of complex fluid simulations with diffusion graph networks. In *International Conference on Learning Representations (ICLR)*, 2025.
- [25] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023.
- [26] Phillip Lippe, Bastiaan S. Veeling, Paris Perdikaris, Richard E. Turner, and Johannes Brandstetter. PDE-Refiner: Achieving accurate long rollouts with neural PDE solvers. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023.
- [27] Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *International Conference on Learning Representations (ICLR)*, 2023.
- [28] James Lucas, George Tucker, Roger Grosse, and Mohammad Norouzi. Don’t blame the ELBO! a linear VAE perspective on posterior collapse. In *Advances in Neural Information Processing Systems 32 (NeurIPS)*, 2019.

- [29] Ziqi Ma, David Pitt, Kamyar Azizzadenesheli, and Anima Anandkumar. Calibrated uncertainty quantification for operator learning via conformal prediction. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856. URL <https://openreview.net/forum?id=cGpegxy12T>.
- [30] Emilia Magnani, Nicholas Krämer, Runa Eschenhagen, Lorenzo Rosasco, and Philipp Hennig. Approximate bayesian neural operators: Uncertainty quantification for parametric PDEs, 2022. URL <https://arxiv.org/abs/2208.01565>.
- [31] Étienne Marcotte, Valentina Zantedeschi, Alexandre Drouin, and Nicolas Chapados. Regions of reliability in the evaluation of multivariate probabilistic forecasts. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 23958–24004. PMLR, 2023.
- [32] Saeed Mohammadzadeh and Emma Lejeune. Predicting mechanically driven full-field quantities of interest with deep learning-based metamodels. *Extreme Mechanics Letters*, 50:101566, 2022. doi: 10.1016/j.eml.2021.101566.
- [33] Roberto Molinaro, Samuel Lanthaler, Bogdan Raonić, Tobias Rohner, Victor Armegoiu, Stephan Simonis, Dana Grund, Yannick Ramic, Zhong Yi Wan, Fei Sha, Siddhartha Mishra, and Leonardo Zepeda-Núñez. Generative AI for fast and accurate statistical computation of fluids, 2024. URL <https://arxiv.org/abs/2409.18359>. Published as B. Raonić et al., “Generative AI for efficient statistical computation of fluids”, *Nature Communications*, 2026, doi: 10.1038/s41467-026-76390-x.
- [34] Lukas Mosser, Olivier Dubrule, and Martin J. Blunt. Reconstruction of three-dimensional porous media using generative adversarial neural networks. *Physical Review E*, 96(4):043309, 2017. doi: 10.1103/PhysRevE.96.043309.
- [35] Ruben Ohana, Michael McCabe, Lucas Meyer, Rudy Morel, Fruzsina J. Agocs, Miguel Beneitez, Marsha Berger, Blakesley Burkhart, Keaton Burns, Stuart B. Dalziel, Drummond B. Fielding, Daniel Fortunato, Jared A. Goldberg, Keiya Hirashima, Yan-Fei Jiang, Rich R. Kerswell, Suryanarayana Maddu, Jonah Miller, Payel Mukhopadhyay, Stefan S. Nixon, Jeff Shen, Romain Watteaux, Bruno Régaldo-Saint Blancard, François Rozet, Liam H. Parker, Miles Cranmer, and Shirley Ho. The well: a large-scale collection of diverse physics simulations for machine learning. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2024), Datasets and Benchmarks Track*, 2024.
- [36] Joel Oskarsson, Tomas Landelius, Marc Peter Deisenroth, and Fredrik Lindsten. Probabilistic weather forecasting with hierarchical graph neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [37] Tobias Pfaff, Meire Fortunato, Alvaro Sanchez-Gonzalez, and Peter W. Battaglia. Learning mesh-based simulation with graph networks. In *International Conference on Learning Representations (ICLR)*, 2021.
- [38] Pierre Pinson and Julija Tastu. Discrimination ability of the energy score. DTU Compute Technical Report-2013 15, Technical University of Denmark, 2013.
- [39] Aram-Alexandre Pooladian, Heli Ben-Hamu, Carles Domingo-Enrich, Brandon Amos, Yaron Lipman, and Ricky T. Q. Chen. Multisample flow matching: Straightening flows with mini-

- batch couplings. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 28100–28127. PMLR, 2023.
- [40] Ilan Price, Alvaro Sanchez-Gonzalez, Ferran Alet, Tom R. Andersson, Andrew El-Kadi, Dominic Masters, Timo Ewalds, Jacklynn Stott, Shakir Mohamed, Peter Battaglia, Remi Lam, and Matthew Willson. Probabilistic weather forecasting with machine learning. *Nature*, 637(8044):84–90, 2025. doi: 10.1038/s41586-024-08252-9.
 - [41] Apostolos F. Psaros, Xuhui Meng, Zongren Zou, Ling Guo, and George Em Karniadakis. Uncertainty quantification in scientific machine learning: Methods, metrics, and comparisons. *Journal of Computational Physics*, 477:111902, 2023. doi: 10.1016/j.jcp.2022.111902.
 - [42] David Ramos, Lucas Lacasa, Fermín Gutiérrez, Eusebio Valero, and Gonzalo Rubio. FluidFlow: A flow-matching generative model for fluid dynamics surrogates on unstructured meshes. arXiv preprint arXiv:2604.08586, 2026.
 - [43] Stephan Rasp, Stephan Hoyer, Alexander Merose, Ian Langmore, Peter Battaglia, Tyler Russell, Alvaro Sanchez-Gonzalez, Vivian Yang, Rob Carver, Shreya Agrawal, Matthew Chantry, Zied Ben Bouallegue, Peter Dueben, Carla Bromberg, Jared Sisk, Luke Barrington, Aaron Bell, and Fei Sha. WeatherBench 2: A benchmark for the next generation of data-driven global weather models. *Journal of Advances in Modeling Earth Systems*, 16(6):e2023MS004019, 2024. doi: 10.1029/2023MS004019.
 - [44] Alvaro Sanchez-Gonzalez, Jonathan Godwin, Tobias Pfaff, Rex Ying, Jure Leskovec, and Peter W. Battaglia. Learning to simulate complex physics with graph networks. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, volume 119 of *Proceedings of Machine Learning Research*. PMLR, 2020.
 - [45] Michael Scheuerer and Thomas M. Hamill. Variogram-based proper scoring rules for probabilistic forecasts of multivariate quantities. *Monthly Weather Review*, 143(4):1321–1334, 2015. doi: 10.1175/MWR-D-14-00269.1.
 - [46] Kihyuk Sohn, Honglak Lee, and Xinchen Yan. Learning structured output representation using deep conditional generative models. In *Advances in Neural Information Processing Systems 28 (NeurIPS)*, 2015.
 - [47] Alexander Tong, Kilian Fatras, Nikolay Malkin, Guillaume Huguet, Yanlei Zhang, Jarrid Rector-Brooks, Guy Wolf, and Yoshua Bengio. Improving and generalizing flow-based generative models with minibatch optimal transport. *Transactions on Machine Learning Research*, 2024.
 - [48] Salvatore Torquato. *Random Heterogeneous Materials: Microstructure and Macroscopic Properties*, volume 16 of *Interdisciplinary Applied Mathematics*. Springer New York, 2002. ISBN 978-1-4757-6357-7. doi: 10.1007/978-1-4757-6355-3.
 - [49] Haixu Wu, Huakun Luo, Haowen Wang, Jianmin Wang, and Mingsheng Long. Transolver: A fast transformer solver for PDEs on general geometries. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pages 53681–53705. PMLR, 2024. Spotlight.
 - [50] Michaël Zamo and Philippe Naveau. Estimation of the continuous ranked probability score with limited information and applications to ensemble weather forecasts. *Mathematical Geosciences*, 50(2):209–234, 2018. doi: 10.1007/s11004-017-9709-7.

- [51] Shengjia Zhao, Jiaming Song, and Stefano Ermon. Infovae: Information maximizing variational autoencoders, 2017. Later published as “InfoVAE: Balancing Learning and Inference in Variational Autoencoders”, AAAI 2019.